

Yueh-Hsuan (Helen) Huang

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EDUCATION

The University of Texas at Austin, Austin, TX

August 2026 – May 2028

Master of Science in Engineering, Electrical and Computer Engineering

Current Coursework: Software Testing, Computer Architecture

National Taiwan University, Taipei, Taiwan

August 2022 – June 2026

Bachelor of Science in Electrical Engineering, GPA 4.12/4.30 (Top 19%)

Additional Program: Business Analytics Program, College of Management

Coursework: Data Structures, Algorithms, Machine Learning, Computer Networks, Introduction to Computers

TECHNICAL SKILLS

- Programming Languages: Python, C, Java, R, MATLAB
- Machine Learning & Data: PyTorch, scikit-learn, NumPy, pandas, Recommendation Systems, Image & Signal Processing
- Cloud & Infrastructure: Google Cloud Platform, Linux, Git, GitHub Actions, Vercel
- Tools & Platforms: Elasticsearch, n8n

EXPERIENCE

TVBS Media Inc., Taipei, Taiwan

July 2025 – March 2026

AI Engineering Intern

- Built an end-to-end news rundown generation workflow that reduced editor-reported manual workload by 75%.
- Integrated n8n, the Google Gemini API, and Elasticsearch into a multimodal news-scoring pipeline that translated a veteran editor's 20 years of editorial judgment into scoring criteria and processed 2,000+ news items monthly.
- Developed and deployed a Google Trends scraper on Google Cloud Run to replace the prior RSS-based approach, increasing keyword-matched news from ~2 to ~20 out of ~80 articles per day.

Interdisciplinary Data Science & Signal Processing Lab, NTU, Taipei, Taiwan

September 2024 – March 2026

Undergraduate Researcher, Advisor: Prof. Che Lin

Topic: Parameter-Efficient Fine-Tuning (PEFT) for Multimodal Sequential Recommendation

- Evaluated LoRA and adapter-based PEFT methods for cross-domain adaptation of the MTSTRec multimodal sequential recommender.
- Fine-tuned only 0.27% of model parameters while maintaining and slightly improving recommendation performance during cross-domain adaptation.
- Funded by the Undergraduate Research Program of Taiwan's National Science and Technology Council (NSTC).

PROJECTS

AI-Powered Product Compliance System for a Leading E-commerce Platform

February 2025 – June 2025

- Led a team of three economics students to build a multimodal compliance-screening system for 4,700+ e-commerce products using LLMs and Google Cloud OCR to identify potential violations of Taiwan's Tobacco Hazards Prevention Act.
- Detected 741 potential violations, 5.3× more than the 141 identified by the official method.
- Prototyped an automated end-to-end n8n workflow that reduced processing time by over 80%.

PUBLICATIONS

- Lin, Y.-C., Chen, Y.-H., Dong, J.-K., **Huang, Y.-H.**, Chen, S.-C., Chen, Y.-C., Chen, C.-Y., Lin, Y.-J., Chen, Y.-L., Chen, Z.-Y., et al. TAU: A Benchmark for Cultural Sound Understanding Beyond Semantics. *IEEE ICASSP 2026*.
- Luo, Y.-X., Lin, Y.-C., Chuang, M.-T., Chen, J.-H., Tsai, I.-N., Kiew, P. X., **Huang, Y.-H.**, Liu, C.-F., Chen, Y.-C., Feng, B.-H., Ren, W., and Lee, H.-Y. ToxicTone: A Mandarin Audio Dataset Annotated for Toxicity and Toxic Utterance Tonality. *Interspeech 2025*.
- Tsai, I.-N., Wu, Y.-X., **Huang, Y.-H.**, Chen, Y.-C., and Ding, J.-J. Optimization of Elevator Standby Scheduling Strategy in Smart Buildings. *Applied System Innovation 2025*.